Intro to Data Science - Lab 8

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Week 8 - Linear Models

```
# Enter your name here: Mars Schrag
```

Please include nice comments.

Instructions:

Run the necessary code on your own instance of R-Studio.

Attribution statement: (choose only one and delete the rest)

```
# 1. I did this lab assignment by myself, with help from the book and the professor.
```

Linear modeling, also referred to as **regression analysis** or multiple regression **bold text**, is a technique for fitting a line, plane, or higher order linear object to data. In their simplest form, linear models have one metric **outcome variable** and one or more **predictor variables** (any combination of metric values, ordered scales such as ratings, or dummy codes).

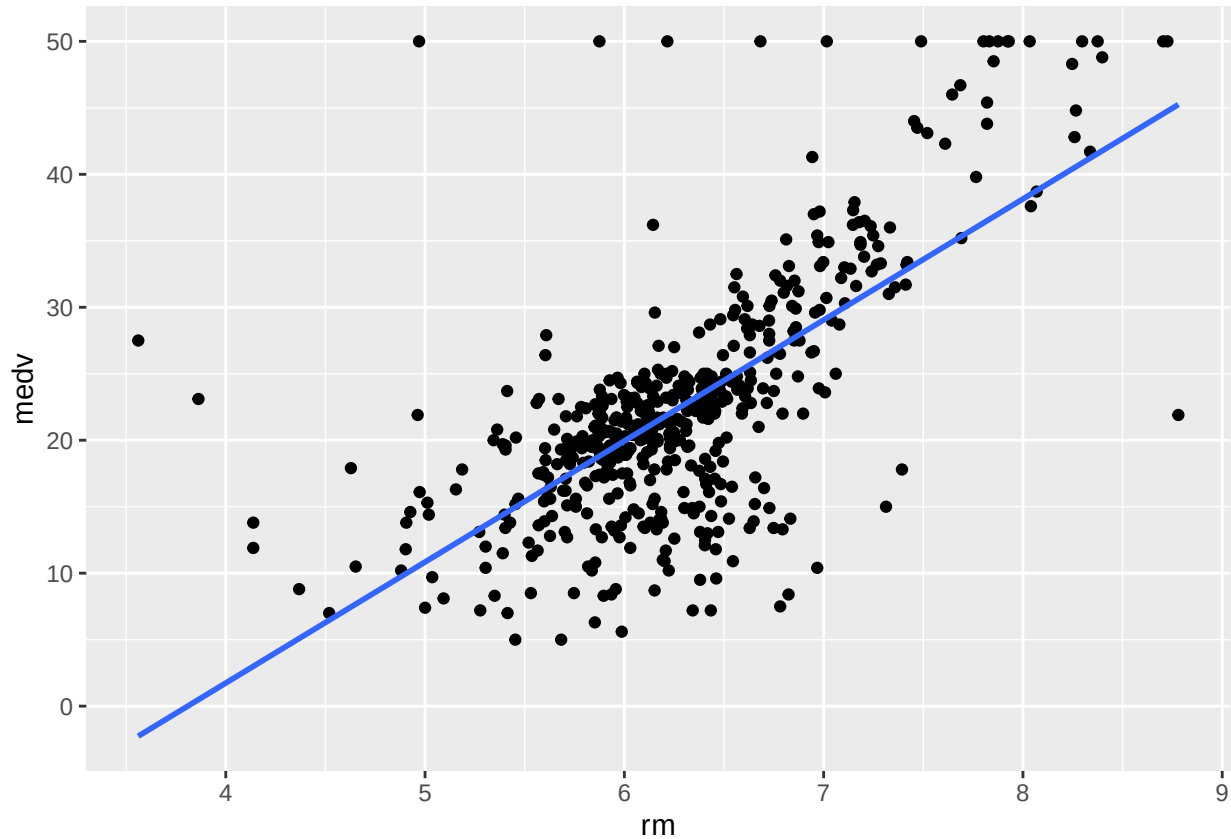
Make sure to library the **MASS** and **ggplot2** packages before running the following:

```
ggplot(data=Boston) + aes(x=rm, y=medv) + geom_point() +
 geom_smooth(method="lm", se=FALSE)
```

```
#install.packages("MASS")
#install.packages("ggplot2")
library(MASS)
library(ggplot2)

ggplot(data=Boston) +
  aes(x=rm, y=medv) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



1. Explore this dataset descrption by typing `?Boston` in a code cell.

```
?Boston
```

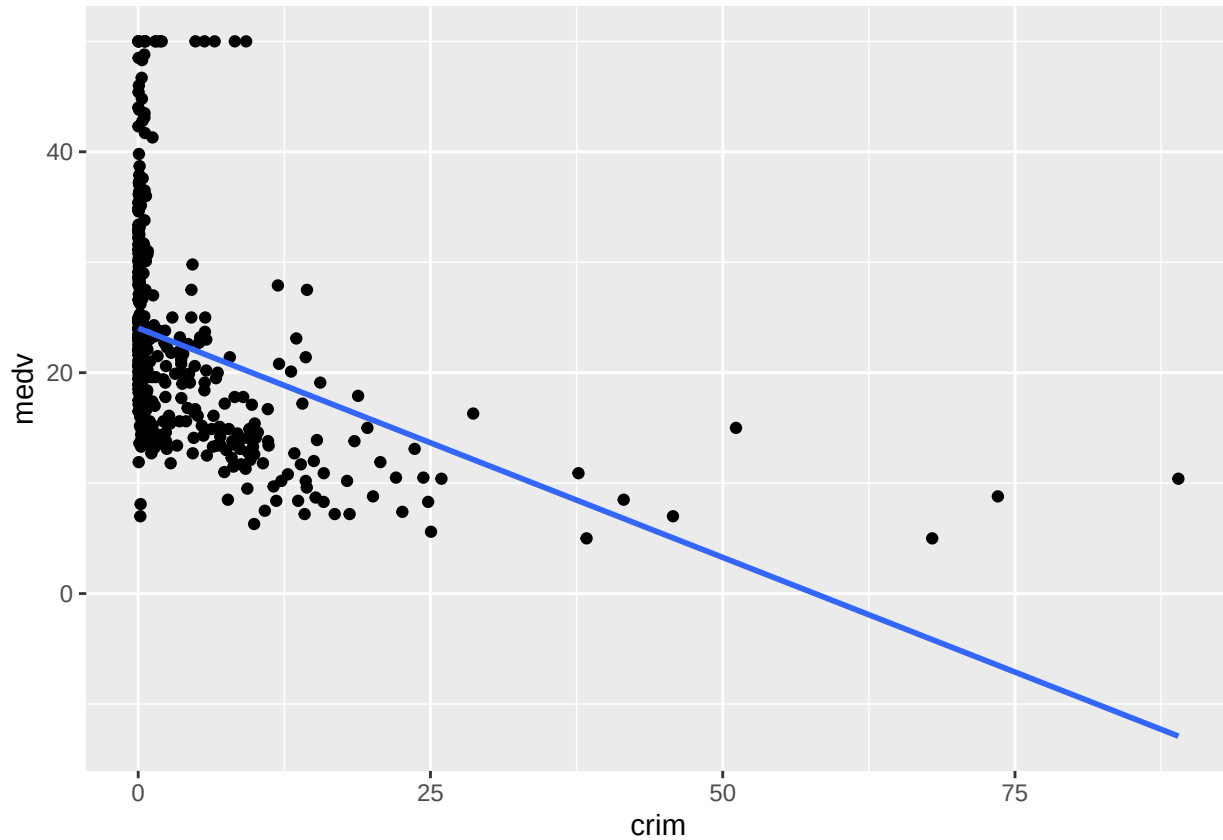
```
## starting httpd help server ... done
```

```
#crim: per capita crime rate by down
#rm: average number of rooms per dwelling
#dis: weighted mean of distances to five boston employment centres
#medv: median value of owner-occupied homes in $1000s
```

2. The graphic you just created fits a best line to a cloud of points. Copy and modify the code to produce a plot where ** crim ** is the x variable instead of ** rm**.

```
ggplot(data=Boston) +
  aes(x=crim, y=medv) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

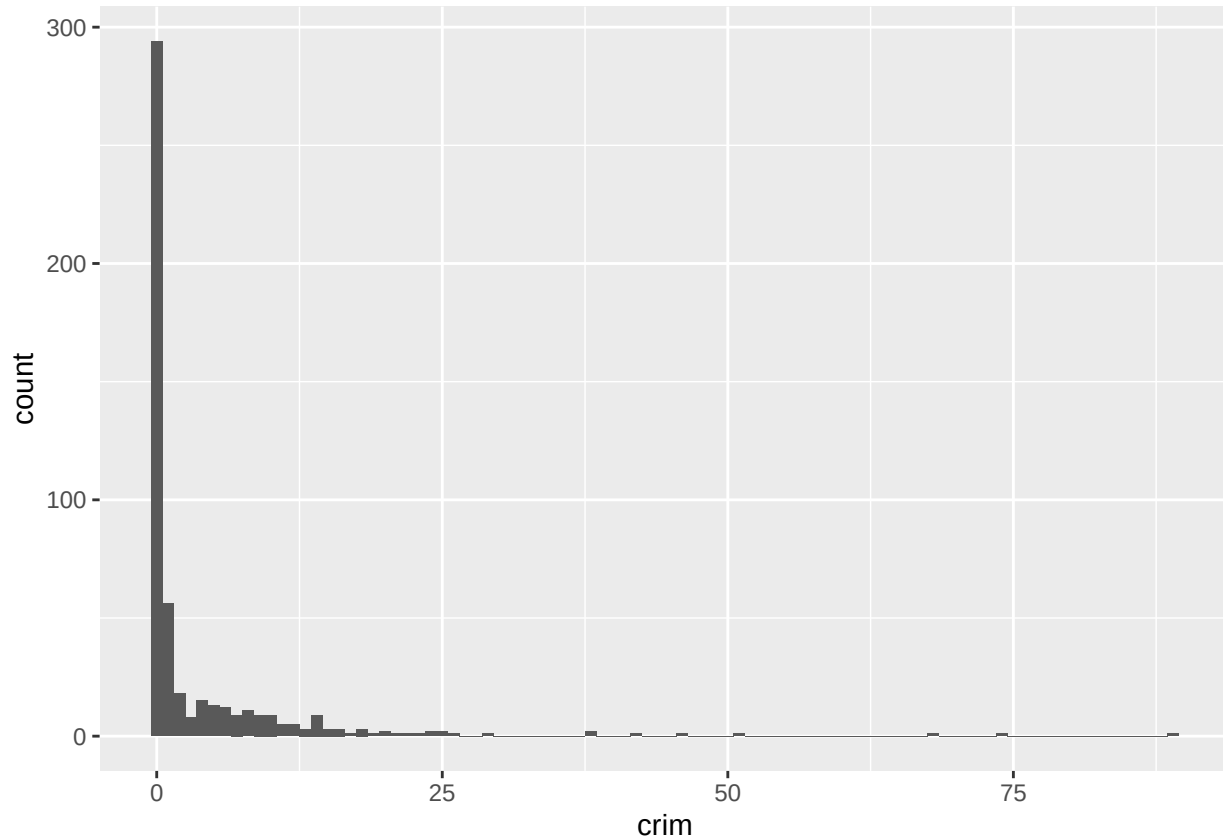


3. Produce a histogram and descriptive statistics for **Boston$crim**. Write a comment describing any anomalies or oddities.

```
summary(Boston$crim)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
##  0.00632  0.08205  0.25651  3.61352  3.67708 88.97620
```

```
ggplot(data = Boston, aes(x=crim))+
  geom_histogram(binwidth = 1)
```



4. Produce a linear model, using the **lm( )** function where **crim** predicts **medv**. Remember that in R s formula language, the **outcome variable** comes first and is separated from the predictors by a **tilde**, like this: `medv ~ crim` Try to get in the habit of storing the output object that is produced by lm and other analysis procedures. For example, I often use **lmOut <- lm(. . .)**

```
lmOut <- lm(medv ~ crim, Boston)
```

5. Run a **multiple regression** where you use **rm**, **crim**, and **dis** (distance to Boston employment centers). You will use all three predictors in one model with this formula: `medv ~ crim + rm + dis` Now run three separate models for each independent variable separate.

```
lmOut_full <- lm(medv~crim+rm+dis, Boston)
summary(lmOut_full)
```

```
##
## Call:
## lm(formula = medv ~ crim + rm + dis, data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -21.247  -2.930  -0.572   2.390  39.072
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -29.45838    2.60010 -11.330  < 2e-16 ***
## crim         -0.25405    0.03532  -7.193 2.32e-12 ***
## rm            8.34257    0.40870  20.413  < 2e-16 ***
## dis           0.12627    0.14382   0.878     0.38
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.238 on 502 degrees of freedom
## Multiple R-squared:  0.5427, Adjusted R-squared:  0.5399
## F-statistic: 198.6 on 3 and 502 DF,  p-value: < 2.2e-16
```

```r
#the p-value tests the null hypothesis: the predictor has no effect
#the p-value represents the probability of observing the given data assuming the null hypothesis is tru

#medv = -29.45838 -0.25405*crim + 8.34257*rm + 0.12627*dis

#r^2: how much of the variation in the dependent variable the model can explain using the independent v
```

```r
lmOut_crim <- lm(medv~crim, Boston)
summary(lmOut_full)
```

```
##
## Call:
## lm(formula = medv ~ crim + rm + dis, data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -21.247  -2.930  -0.572   2.390  39.072
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -29.45838    2.60010 -11.330  < 2e-16 ***
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## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.238 on 502 degrees of freedom
## Multiple R-squared:  0.5427, Adjusted R-squared:  0.5399
## F-statistic: 198.6 on 3 and 502 DF,  p-value: < 2.2e-16
```

```r
lmOut_rm <- lm(medv~rm, Boston)
summary(lmOut_full)
```

```
##
## Call:
## lm(formula = medv ~ crim + rm + dis, data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -21.247  -2.930  -0.572   2.390  39.072
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -29.45838    2.60010 -11.330  < 2e-16 ***
## crim         -0.25405    0.03532  -7.193 2.32e-12 ***
## rm            8.34257    0.40870  20.413  < 2e-16 ***
## dis           0.12627    0.14382   0.878     0.38
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.238 on 502 degrees of freedom
## Multiple R-squared:  0.5427, Adjusted R-squared:  0.5399
## F-statistic: 198.6 on 3 and 502 DF,  p-value: < 2.2e-16
```

```r
lmOut_dis <- lm(medv~dis, Boston)
summary(lmOut_full)
```

```
##
## Call:
## lm(formula = medv ~ crim + rm + dis, data = Boston)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -21.247  -2.930  -0.572   2.390  39.072
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -29.45838    2.60010 -11.330  < 2e-16 ***
## crim         -0.25405    0.03532  -7.193 2.32e-12 ***
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## dis           0.12627    0.14382   0.878     0.38
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.238 on 502 degrees of freedom
## Multiple R-squared:  0.5427, Adjusted R-squared:  0.5399
## F-statistic: 198.6 on 3 and 502 DF,  p-value: < 2.2e-16
```

6. Interpret the results of your analysis in a comment. Make sure to mention the **p-value**, the **adjusted R-squared**, the list of **significant predictors** and the **coefficient** for each significant predictor.

```r
# according to the adjusted r^2, 53.99% of the variation in medv is explained by crim, rm, and dis
# a predictor is statistically significant if its p-value < 0.05. crim and rm affect house prices, but
```

7. Create a one-row **data frame** that contains some plausible values for the predictors. For example, this data frame contains the median values for each predictor: `predDF <- data.frame(crim = 0.26, dis=3.2, rm=6.2)` The numbers used here were selected randomly by looking at min and max data of the variables.

```r
predDF <- data.frame(crim = 0.26, dis=3.2, rm=6.2)
predDF
```

```
##   crim dis  rm
## 1 0.26 3.2 6.2
```

8. Use the **predict( )** command to predict a new value of **medv** from the one-row data frame. If you stored the output of your lm model in **lmOut**, the command would look like this: `predict(lmOut, predDF)`

```r
predict(lmOut_full, predDF)
```

```
##        1
## 22.60355
```